

Math 170A - Homework 1 - due October 11

Unless specified otherwise, show all your work.

Problem 1. Let A and B be two events. If $\mathbb{P}(A \cup B) = 0.9$, $\mathbb{P}(A^c \cup B) = 0.8$ and $\mathbb{P}(B^c \cup A) = 0.4$, what is $\mathbb{P}(A \cap B)$?

Problem 2. Let A, B, C be sets and $\mathbb{P}$ be a probability law.

Which of the following must be true? Prove or give a counterexample:

- a) $A \subset (A \cap B) \cup B^c$
- b) $(A \cup B^c) \cap B = \emptyset$
- c) $(A \cap B)^c \cap C = (A^c \cap C) \cup (B^c \cap C)$
- d) $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B) + \mathbb{P}(A^c \cap B^c \cap C)$

(Hint: Draw Venn's diagrams first to see if a statement looks true. If yes, then prove it by definition(s). If no, think of small finite sets that give counterexample.)

Problem 3. Suppose that A and B are two events such that $0 < \mathbb{P}(B) < 1$, $\mathbb{P}(A|B) \geq 0.5$ and $\mathbb{P}(A|B^c) \geq 0.5$. Show that $\mathbb{P}(A) \geq 0.5$.

Problem 4. Alice and Bob people arrive at the train station sometime between 4:00 and 4:20 with uniform probability and independent of each other. The trains arrive at 4:02, 4:12, 4:22.

- (i) What is the probability that Alice needs to wait more than 2 minutes?
- (ii) What is the probability both of these two people take the 4:02 train?
- (iii) What is the probability that these two people take different trains?
- (iv) What is the probability that they both take the 4:12 train given that they take the same train?

Problem 5. A store sells three types of shoes: running shoes (40% of all shoes in the store), hiking shoes (30%) and raining shoes (30%). It is known that 10% of running shoes are water proof; 70% of hiking shoes are water proof; all raining shoes are, of course, water proof. If a selected shoe turns out to be water proof, what is the probability that it is a hiking shoe?

Problem 6. You read about a survey in the newspaper, which says the following: *"If you're driving a compact car, the probability you survive the collision with a sport car is 0.9, and the probability you survive the collision with an SUV is 0.85. If you're driving a sport car the probability you survive a collision with a compact car is 0.8 and the the probability you survive a collision with a SUV is 0.7. If you're driving an SUV the probability you survive a collision with a compact car is 0.95, and the collision with a sport car is 0.9. Whatever, you're driving the probability you survive a crash with a car of the same type (compact, sport or SUV) is 0.8"*

You just moved into a new city in which 20% of cars are compact, 20% of cars are sport cars and 60% of cars are SUVs, and you want to buy a car to use it in the city. What car would you buy to maximize the chance of surviving in a collision? Assume that collision of any pair of cars has the same probability.

(Disclaimer: all the numbers in this problem are completely made up.)

Problem 7. We have a very weird die, for which we know that the probability that the number is strictly less than 5 is 0.6, the probability that we get 4 is equal to 0.2 and the probability that the number is strictly bigger than 2 is 0.9. Show that this information is not sufficient to determine the probability law, that is there is more than one probability law for this experiment which has the above properties.